



Inversion for solar rotation rate using seismic data

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Abstract

The subject of helioseismology aims to infer the solar structure and dynamics through analysis of the observed normal mode oscillations. Rotational splittings of frequencies of normal modes can be used to infer the rotation rate in the solar interior as a function of radius and latitude. Temporal variations in the rotation rate are studied using GONG data for the last 25 years covering two solar cycles.

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I. INTRODUCTION

The Sun is an impossibly impenetrable object, offering very little by way of experimentation and resulting study. In fact, in the words of Sir Arthur Eddington, [1] “At first sight, it would seem that the deep interior of the Sun and stars is less accessible for scientific investigation than any other region of the universe. Our telescopes may probe farther and farther into the depths of space, but how can we ever obtain certain knowledge of that which is hidden behind substantial barrier? What appliance can pierce through the outer layers of a star and test the conditions within?”. Hence, the difficulty of pursuing knowledge of the interior structure, and conditions of the Sun, is evident. The field of helioseismology attempts to do precisely that. One of the sources on information that are available to us, is the oscillation frequencies of the Sun. Using the observed oscillations in solar radiation, the velocity field beneath the surface layer is calculated. The temporal variation of the thus calculated velocity field, is then studied. The inversion of solar oscillations frequencies have been studied extensively, the more recent studies on inversions for rotation rate in solar interior, being Schou et al. 1998 [2] and by Antia et al. 2008[3]. The observed

data, ranging over the solar cycles 23 and 24 have been studied, and the correlations between the occurrence of sunspots, and the velocity of the matter beneath it, have been observed.

II. BASIC EQUATIONS

To provide a background for the oscillatory equations, and their analysis, it is important that we clarify the physical equations governing the stellar oscillations.

i. The equation of continuity

The conservation of mass can be written as, [4]

$$\frac{\partial \rho}{\partial t} + \text{div}(\rho \mathbf{v}) = 0, \quad (1)$$

where ρ is the mass density, $\mathbf{v}$ is the velocity, and t represents time. This may also be written as the continuity equation,

$$\frac{d\rho}{dt} + \rho \text{div} \mathbf{v} = 0, \quad (2)$$

where $\frac{d}{dt}$ is the material derivative, defined as

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + \mathbf{v} \cdot \nabla \phi; \quad (3)$$

where ϕ is a scalar field

ii. The equations of motion

Under stellar oscillations, the viscosity is neglected, under the adiabatic approximation. Hence, the only forces acting on a volume of gas are,[4]

- Surface forces, i.e, pressure on the surface
- Body forces

with these considerations, the equations of motion may be written as

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \rho \mathbf{g}, \quad (4)$$

where $\mathbf{g}$ is the acceleration due to gravity, which is the only body force which is considered.

It should also be noted that if $\mathbf{g}$ is the gradient of a potential(ϕ), then it must satisfy Poisson's equation

$$\nabla^2 \phi = 4\pi G \rho. \quad (5)$$

iii. The energy equation

In the adiabatic approximation, the time scale of oscillations is considered to be small enough, that there is no exchange of energy between fluid elements, in the relevant time scales[4] In the adiabatic approximation, ρ and p are related as:

$$\frac{dp}{dt} = \frac{\Gamma_1 p}{\rho} \frac{d\rho}{dt}, \quad (6)$$

where Γ_1 is the adiabatic index, defined:

$$\Gamma_1 = \left(\frac{\partial \ln p}{\partial \ln \rho} \right)_{ad}. \quad (7)$$

iv. Perturbation analysis

For the study of oscillations, we will consider small perturbations around the equilibrium state. We shall consider two kinds of perturbative analysis,[4]

- the Eulerian perturbation, in which the coordinates under observation are fixed, while the physical parameter varies
- the Lagrangian perturbation, in which the particle moves in space, and the observation follows the motion of the particle.

To make this difference distinct, we write the Eulerian perturbation to the pressure as,

$$p(\mathbf{r}, t) = p_0(\mathbf{r}) + p'(\mathbf{r}, t), \quad (8)$$

where p_0 is the equilibrium pressure, and p' is the Eulerian perturbation.

The Lagrangian perturbation may be written as

$$\delta p(\mathbf{r}) = p(\mathbf{r}_0 + \delta \mathbf{r}) - p_0(\mathbf{r}_0) = p(\mathbf{r}_0) + \delta \mathbf{r} \cdot \nabla p_0 - p_0(\mathbf{r}_0) = p'(\mathbf{r}_0) + \delta \mathbf{r} \cdot \nabla p_0. \quad (9)$$

In the steady state the time derivative and velocity vanish and the equilibrium quantities satisfy the equation of hydrostatic equilibrium. Assuming the perturbations to be small and neglecting higher order terms we are left with linear equations in the perturbations, which may be written as,

$$\frac{\partial \rho'}{\partial t} + \text{div}(\rho_0 \mathbf{v}) = 0, \quad (10)$$

$$\rho_0 \frac{\partial^2 \delta \mathbf{r}}{\partial t^2} = \rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\nabla p' + \rho_0 \mathbf{g}' + \rho' \mathbf{g}_0, \quad (11)$$

$$\nabla^2 \phi' = 4\pi G \rho', \quad (12)$$

$$\frac{\partial p'}{\partial t} - \frac{\Gamma_1 p_0}{\rho_0} \frac{\partial \rho'}{\partial t} = 0. \quad (13)$$

v. Adiabatic oscillations

The displacement due to the oscillations $\delta \mathbf{r}$ may be separated into radial and horizontal components as[4]

$$\delta \mathbf{r} = \xi_r \mathbf{a}_r + \xi_h. \quad (14)$$

The horizontal component of equation 10 may be written as

$$\rho_0 \frac{\partial^2 \xi_h}{\partial t^2} = -\nabla_h p' - \rho_0 \nabla_h \phi', \quad (15)$$

and the radial component is

$$\rho_0 \frac{\partial^2 \xi_r}{\partial t^2} = -\frac{\partial p'}{\partial r} - \rho' g_0 - \rho_0 \frac{\partial \phi'}{\partial r}. \quad (16)$$

Taking the horizontal component of the divergence of equation 15, we get

$$\rho_0 \frac{\partial^2}{\partial t^2} \nabla_h \cdot \xi_h = -\nabla_h^2 p' - \rho_0 \nabla_h^2 \phi'. \quad (17)$$

We write the equation of continuity in the following form

$$\rho' = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho_0 \xi_r) - \rho_0 \nabla_h \cdot \xi_h. \quad (18)$$

Using this, we eliminate $\nabla_h \cdot \xi_h$ from equation 17, which becomes

$$-\frac{\partial^2}{\partial t^2} \left[\rho' + \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \rho_0 \xi_r) \right] = -\nabla_h^2 p' - \rho_0 \nabla_h^2 \phi'. \quad (19)$$

The equations of the horizontal component of the laplacian may be solved explicitly, to obtain the eigenfunctions to be the spherical harmonics $Y_l^m(\theta, \phi)$, with eigenvalue $-\frac{l(l+1)}{r^2}$. The angular part of the dependent variables in (14)-(17) may be solved as

$$\xi_r(r, \theta, \phi, t) = \sqrt{4\pi} \tilde{\xi}_r(r) Y_l^m(\theta, \phi) \exp(-i\omega t), \quad (20)$$

$$p'(r, \theta, \phi, t) = \sqrt{4\pi} \tilde{p}'(r) Y_l^m(\theta, \phi) \exp(-i\omega t), \quad (21)$$

where $\tilde{\xi}$ and $\tilde{p}'$ represent the amplitude functions of those quantities. The horizontal component of the displacement may be solved independently, from (15) to obtain

$$\xi_h(r, \theta, \phi, t) = \sqrt{4\pi} \tilde{\xi}_h(r) \left(\frac{Y_l^m(\theta, \phi)}{\partial \theta} \mathbf{a}_\theta + \frac{1}{\sin \theta} \frac{Y_l^m(\theta, \phi)}{\partial \phi} \mathbf{a}_\phi \right) \exp(-i\omega t). \quad (22)$$

Using these solutions in the differential equations, and dividing by the common factor of $Y_l^m(\theta, \phi) \exp(-i\omega t)$, we obtain the equations for the radial part of the oscillations:

$$\omega^2 \left[\tilde{\rho}' + \frac{1}{r^2} \frac{d}{dr} (r^2 \rho_0 \tilde{\xi}_r) \right] = \frac{l(l+1)}{r^2} (\tilde{p}' + \rho_0 \tilde{\phi}'), \quad (23)$$

$$-\omega^2 \rho_0 \tilde{\xi}_r = -\frac{d\tilde{p}'}{dr} - \tilde{\rho}' g_0 - \rho_0 \frac{d\tilde{\phi}'}{dr}, \quad (24)$$

and Poisson's equation may also be similarly rewritten as

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\tilde{\phi}'}{dr} \right) - \frac{l(l+1)}{r^2} \tilde{\phi}' = 4\pi G \tilde{\rho}'. \quad (25)$$

The energy equation must now be dealt with independently. Under the adiabatic approximation, it is written in the form

$$\rho' = \frac{\rho}{\Gamma_1 p} p' + \rho \tilde{\xi}_r \left(\frac{1}{\Gamma_1 p} \frac{dp}{dr} - \frac{1}{\rho} \frac{d\rho}{dr} \right). \quad (26)$$

This may be used to eliminate ρ' from the previous differential equations. Dropping the subscript '0' for equilibrium quantities, and the tilde for amplitude functions, we write the oscillation equations as

$$\frac{d\tilde{\xi}_r}{dr} = -\left(\frac{2}{r} + \frac{1}{\Gamma_1 p} \frac{dp}{dr} \right) \tilde{\xi}_r + \frac{1}{\rho c^2} \left(\frac{S_l^2}{\omega^2} - 1 \right) p' + \frac{l(l+1)}{\omega^2 r^2} \phi', \quad (27)$$

$$\frac{dp'}{dr} = \rho(\omega^2 - N^2) \tilde{\xi}_r + \frac{1}{\Gamma_1 p} \frac{dp}{dr} p' - \rho \frac{d\phi'}{dr}, \quad (28)$$

where N is the buoyancy frequency, defined as

$$N^2 = g \left(\frac{1}{\Gamma_1 p} \frac{dp}{dr} - \frac{1}{\rho} \frac{d\rho}{dr} \right), \quad (29)$$

and S_l is the characteristic acoustic frequency of the modes, defined as

$$S_l^2 = \frac{l(l+1)c^2}{r^2} = k_h^2 c^2. \quad (30)$$

These oscillation equations completely describe the physics of the system. To analyse them analytically, we shall later introduce certain approximations.

vi. Boundary conditions

To supplement the fourth order differential equations, four boundary conditions are required. They can be summarised as follows[4]: The centre, $r = 0$, is a singular point of the equations. Two conditions serve to admit only the regular solutions, at the centre. By expanding the solutions, it can be seen that near $r = 0$, ξ_r behaves as r^{l-1} , whereas p' and ϕ' behave as r^l . From the expansions, two relations between the solution near $r=0$ may be obtained, and for $l > 0$,

$$\xi_r \simeq l \xi_h, \quad r \rightarrow 0. \quad (31)$$

The second boundary condition may be obtained by demanding $\phi' \sim r^l$, and may be written as

$$\frac{d\phi'}{dr} = \left(\frac{l}{r}\right) \phi', \quad r \rightarrow 0. \quad (32)$$

One surface condition is obtained by demanding continuity of ϕ' and its derivative at the surface radius $r = R$. Outside the star the density perturbation vanishes, and Poisson's equation may be solved analytically. The solution vanishing at infinity is

$$\phi' = A r^{-l-1}, \quad (33)$$

where A is a constant. Hence, the boundary condition becomes

$$\frac{d\phi'}{dr} + \frac{l+1}{r} \phi' = 0, \quad r = R. \quad (34)$$

For the second surface condition, it is sufficient to take the Lagrangian pressure perturbation to vanish, giving

$$\delta p = p' + \xi_r \frac{dp}{dr} = 0, \quad r = R. \quad (35)$$

III. PHYSICAL NATURE OF THE MODES OF OSCILLATION

The equations 27–30, as defined above, form the complete description of oscillations. However, this general equation is of fourth order, and the solutions can only be obtained numerically. In the case, it appears sensible to approximate them, to a point, where their solutions may be classified and discussed.

i. The Cowling approximation

The Cowling approximation ignores the perturbation to the gravitational potential. This is valid under the considerations of large degree (l) of the mode, or large radial order ($|n|$). However, it must be mentioned that this approximation is not fully justified, outside this region of large l or $|n|$, and hence, the conclusions must be regarded with some caution.

Under the Cowling approximation, we can take the equations governing the oscillations to be the following:[4]

$$\frac{d\xi_r}{dr} = - \left(\frac{2}{r} - \frac{1}{\Gamma_1} H_p^{-1} \right) \xi_r + \frac{1}{\rho c^2} \left(\frac{S_l^2}{\omega^2} - 1 \right) p', \quad (36)$$

and

$$\frac{dp'}{dr} = \rho(\omega^2 - N^2)\xi_r - \frac{1}{\Gamma_1} H_p^{-1} p', \quad (37)$$

where

$$H_p^{-1} = -\frac{d \ln p}{dr}, \quad (38)$$

is the pressure scale height. For oscillations at a higher radial order, the derivatives of the equilibrium quantities are very small, and hence, are neglected. This gives

$$\frac{d\xi_r}{dr} = \frac{1}{\rho c^2} \left(\frac{S_l^2}{\omega^2} - 1 \right) p', \quad (39)$$

and

$$\frac{dp'}{dr} = \rho(\omega^2 - N^2)\xi_r. \quad (40)$$

Combining them, and neglecting the derivatives of basic quantities, such as ρ , N , S_l etc, we can write a second order equation for the oscillations, as

$$\frac{d^2 \xi_r}{dr^2} = -K(r)\xi_r, \quad (41)$$

where $K(r)$ is given by

$$K(r) = \frac{\omega^2}{c^2} \left(\frac{N^2}{\omega^2} - 1 \right) \left(\frac{S_l^2}{\omega^2} - 1 \right). \quad (42)$$

The solution of which is of the form

$$\xi_r \sim \cos\left(\int K^{1/2} dr + \alpha\right), K(r) > 0. \quad (43)$$

It is clear that the solution is oscillatory in the region where $K(r) > 0$, and exponential in case $K(r) < 0$. Usually there is an oscillatory region between two regions where the mode decays exponentially. In such a condition, the mode is said to be trapped between the evanescent regions.

Here, it can be seen to oscillate when either

$|\omega| < N$ and $|\omega| < S_l$ or

$|\omega| > N$ and $|\omega| > S_l$

In the former case, the pressure gradient determines the nature of oscillation of the modes, and hence are labelled as p modes, and in the latter, the buoyancy drives the oscillations, and they are, labelled as g modes.

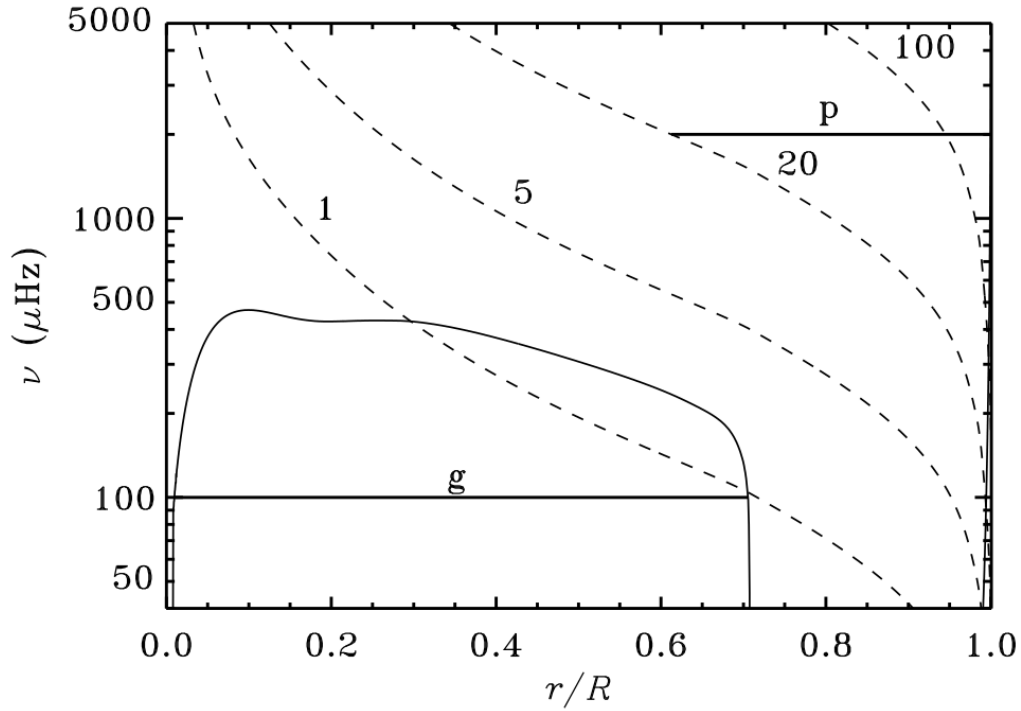


Figure 1: Variation of N and S_l with radial distance. The solid line represents the variation of N , and the dotted lines represent S_l 's for each labelled l . The S_l has a specific l dependence, and hence, the p modes for a specific frequency can only penetrate from the surface until it reaches the dotted line corresponding to its l value. g modes are low frequency modes which are trapped within the boundary of the solid line, and its S_l frequency.[4]

ii. P modes

P modes are high frequency modes, where the frequency of oscillation is higher than both the buoyancy frequency, and the acoustic frequency. Because of this property, the variation on the RHS of equation 39 comes from the p' term. Hence, the variation in pressure determines the eigenfunctions, and hence the nature of the modes. Hence, these are referred to as p modes. From figure 1, we can see, that the p mode oscillations can penetrate only up to where the frequency is greater than the acoustic and buoyancy frequency of that particular mode. This traps the mode between the surface, and an inner turning point, $r = r_t$, where

$$\frac{c^2(r_t)}{r_t^2} = \frac{\omega^2}{l(l+1)}, \quad (44)$$

which fixes $r_t(l, \omega)$. From the nature of the dispersion relation, we can identify $K(r)$, of the p mode to the radial part of the wave vector. This implies that the acoustic waves bend away from the normal as it reaches higher sound speeds. This is identical to the refraction of light, in a medium of varying refractive index. Also, in this approximation, $K(r)$ increases with ω . Hence, the solution to the eigenfunction equation, also oscillates with increasing frequency. This implies that the number of radial nodes of the eigenfunction increases with increasing frequency ω .

iii. G modes

G modes are low frequency modes, where the turning points are determined by $N = \omega$. Here the $K(r)$ may be approximated to $\frac{1}{\omega^2}(N^2 - \omega^2)\frac{l(l+1)}{r^2}$. Thus K increases on decreasing ω , and hence, increase in frequency of the mode, leads to a reduction in radial order. In this case, gravity, through N , provides the normal restoring force. Hence, this mode is called the g mode. Here we find an upper limit of ω , which it cannot cross, and hence, we can see the mode being trapped in the interior of the Sun, between the two trapping points where $r_t(\omega = N)$.

The g modes, after passing through the evanescent region, cross the acoustic frequency at a point before reaching the surface. Therefore, from that point on till the surface, it is identified as a p mode.

iv. Radial order of a mode

The nature of the nodes may be quantified by observing the phase diagram for equation 39 and 40. Plotting p' along the y axis and ξ_r along the x axis, we find that a p mode turns counter clockwise, while a g mode turns clockwise about the origin. Hence, defining the radial order as the number of nodes in the anticlockwise direction, we assign a sign convention which assigns a positive number of nodes for p modes, and negative nodes for g modes. There are certain mixed modes as well, which act as both p and g modes in their respective regions, and in that case, the clockwise and anticlockwise nodes cancel out, and we are left with an effective n.

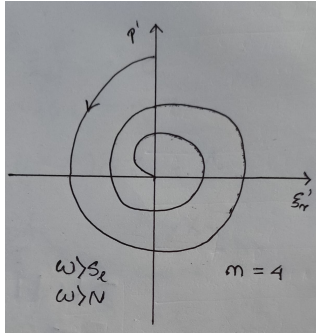


Figure 2: Schematic diagram of counting the number of nodes for a p mode

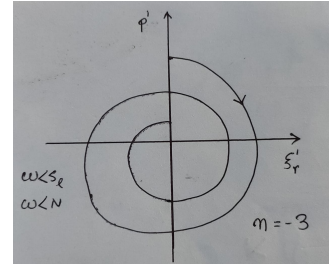


Figure 3: Schematic diagram of counting the number of nodes for a g mode

IV. THE EFFECTS OF ROTATION AND STELLAR OSCILLATIONS

As we have used the spherical harmonic functions to describe the equilibrium model of the non rotating system, it can be shown that the introduction of a constant rotation Ω of the system, would alter the rotating coordinates from (r, θ, ϕ) to (r', θ', ϕ') , where the transformation equation is given by[4]

$$(r', \theta', \phi') = (r, \theta, \phi - \Omega t), \quad (45)$$

where Ω is the angular rotation rate about the polar axis. Translating to spherical harmonics, it can be seen that in the inertial frame, the m dependence changes, from $\cos(m\phi - \omega_0 t)$ to $\cos(m\phi - \omega_m t)$ where

$$\omega_m = \omega_0 + m\Omega. \quad (46)$$

Thus, an observer in an inertial frame finds that the frequency is split uniformly according to m .

i. The Inverse Problem

Now, we may deduce, that the observed frequencies may be split into two parts, the m averaged part ω_{nl} , and a perturbation $\delta\omega_{nl}(m)$ which depends on m . This perturbation, comes from the variation of the velocity of the source, and hence, they can be used to evaluate the velocity field. This difference between m dependent and the m independent part of the observed oscillations may be expanded in terms of any linear combination of basis functions, however, it is convenient to use the Wigner 3- j symbols for this purpose. The expansion of ω , as used here, is [5]

$$\omega_k^m = \omega_k + \delta\omega_m = \omega_{nl} + \sum_{s=1,3,5..} \gamma_{sl}^m c_{sl}^n, \quad (47)$$

where k is the index specifying n and l , and γ is the corresponding Wigner 3- j symbols. The advantage of this expansion is that, firstly, the basis functions are now orthogonal over a discrete field. Secondly, the assumption being used here is the isolated multiplet approximation, i.e, the eigenvalues of the modes with same n and l , constituting a multiplet are distinct from other such multiplets. Under this approximation, the frequency splitting coefficients, c_s are nonzero only if

- The flow is toroidal
- s is odd.

It is to be noted that the c_s for even s is zero only when the second order perturbations are neglected. The main advantage of using the splitting coefficients is that they reduce the dimension of the problem. Instead of having $2l + 1, m$ values, there are only a small number of significant s values, and the effect of each term is very small for each increasing s .

Using variational principle, and treating the rotation term as a perturbation to the operator of adiabatic oscillations, we find the perturbation in the eigenvalues to be given by the integral equation [5]

$$c_{sl}^n = \int_0^{R_\odot} w_s^0(r)_n \mathcal{K}_{ls}(r) r^2 dr, \quad (48)$$

where $\mathcal{K}$, the kernel of the problem, is defined as

$${}_n\mathcal{K}_{ls}(r) = -\rho_0 r^{-1} \left(\tilde{\zeta}_r^2 + L^2 \tilde{\zeta}_h^2 - [2\tilde{\zeta}_r \tilde{\zeta}_h + \frac{1}{2}s(s+1)\tilde{\zeta}_h^2] \right) / \mathcal{N}, \quad (49)$$

Where $\mathcal{N}$ is a scalar which depends on the normalisation of $\tilde{\zeta}_r$ and $\tilde{\zeta}_h$, and is given by

$$\mathcal{N} = \int_0^{R_\odot} \rho_0 (\tilde{\zeta}_r^2 + L^2 \tilde{\zeta}_h^2) r^2 dr \quad (50)$$

where $L = \sqrt{l(l+1)}$, and

$$\mathbf{v}_{rot}(r, \theta) = \sum_{s=1,3,5..}^{\infty} w_s^0(r) \partial_\theta Y_s^0 \hat{\phi}, \quad (51)$$

where $\tilde{\zeta}_r(r)$ and $\tilde{\zeta}_h(r)$ denote, respectively, the radial and horizontal eigenfunctions for harmonic degree l and radial order n .

ii. The RLS Inversion Technique

In order to invert the integral equation, the technique of Regularised Least Squares (RLS) was used. The unknown velocity fields is first expanded in terms of B-spline basis function. Then,

the coefficients of the B-splines are evaluated by solving the system of linear equations for the coefficients.

$$c_i = \sum_{j=1}^N a_j \int_0^{R_\odot} \mathcal{K}_i(r) \phi_j(r) r^2 dr = \sum_{j=1}^N b_{ij} a_j, \quad (52)$$

where

$$w_s^0 = \sum_j a_j \phi_j, \quad (53)$$

where ϕ_j is the j^{th} B-spline basis function, c_i is the splitting coefficient for the i^{th} mode. The $\mathcal{K}_i$, is the i^{th} kernel, defined previously.

However, the straightforward least squares solution will not show any meaningful result, because the solutions will show small scale oscillations, which may not be physically sensible. In that case, the regularisation is done by minimising either the first, or the second derivative, to reduce the oscillation. For first derivative regularisation, the following function is minimised[6]:

$$\sum_{i=1}^n \left(\frac{c_i - \sum_{j=1}^N b_{ij} a_j}{\epsilon_i} \right)^2 + \alpha \sum_{i=1}^M w(r_i) \left(\sum_{j=1}^N a_j \frac{d\phi_j}{dr}(r_i) \right)^2, \quad (54)$$

where α is the regularisation parameter, and w is a weight function, which is to be included as needed. $i = 1, 2, \dots M$ runs over a mesh which suitably covers the range, $j = 1, 2, \dots N$ runs over the range of B-spline basis functions, and $i = 1, 2, \dots n$ runs over the number of modes for which splitting data is available. In practice, the minimisation problem is solved using the SVD. SVD, or the Singular Value Decomposition of a matrix is the process of factoring a rectangular matrix A into a diagonal matrix(Σ), an orthogonal matrix(V), and the matrix of normalised eigenvectors(U), as $A = U\Sigma V^T$. A in this case, is an $(n + M, N)$ matrix, where the first n rows are the $\frac{b_{ij}}{\sigma_i}$, and the remaining M rows are the $\alpha w(r_i) \frac{d^2 \phi_j}{dr^2}(r_i)$. Once the singular values Σ are obtained, the coefficients a_j are evaluated by solving the set of linear equation $Aa = d$, where a is the vector with components, a_j and $d_i = c_i / \sigma_i$ for $i \leq n$ with the remaining M entries set to 0. It is a very powerful tool for obtaining the least squares solution of overdetermined systems of linear equations.

It should be noted that $\alpha = 0$ reduces the equation to solving the least squares solution, and larger the value of α , the less oscillations will be present in the solution. Of course, if the true solution contains oscillations, that will be damped down as well. The ideal value of α cannot be obtained via calculation, and depends on the kind of data, and the problem that is being solved. The advantage of using regularisation is that for the regions where the splitting coefficients c_i are not enough to determine the required solution, it results in a smoother solution, which is more likely to be correct. All the terms from equation 48 are known, in the inversion problem, apart from w_s^0 . The inverse problem is to solve the integral equation and evaluate w_s^0 , and hence, obtain the rotational velocity field.

V. DATA AND CALCULATIONS

The model file for the calculated value for splittings is sninvbb77. This model in particular, has been constructed to match the sound speed and density profile of the Sun. The observed frequencies from the Helioseismic and Magnetic Imager(HMI) were used as the observed data. The difference in frequencies between the observed frequency and model frequency were first plotted as a function of the frequency (in μHz).

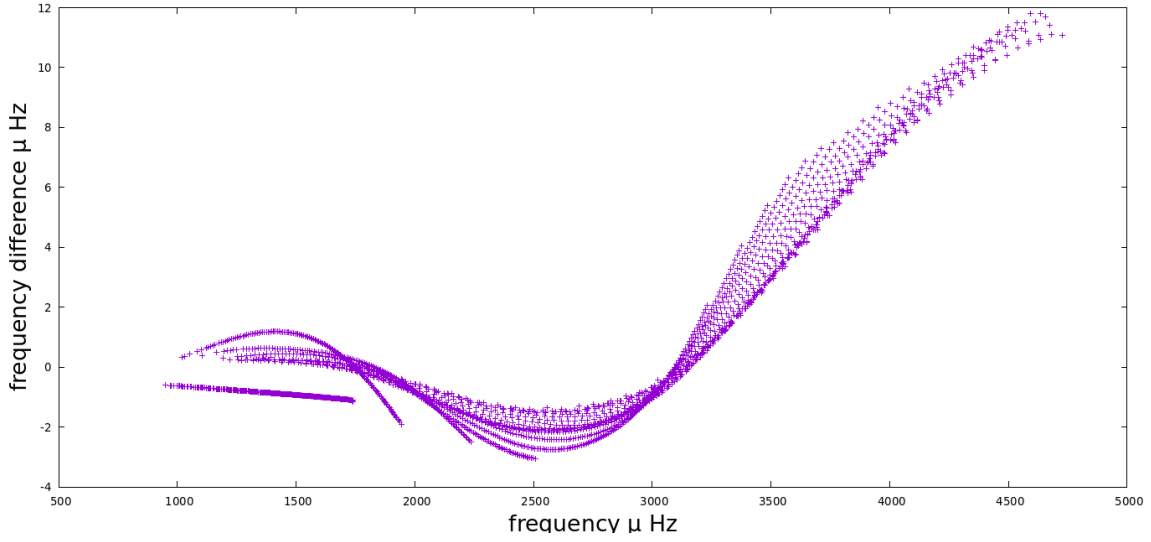


Figure 4: Variation of δv_{nl} with v_{nl}

The nonzero frequency difference is different for different modes. Now, defining the energy of each mode as

$$E_{nl} = 4\pi \frac{\int_0^R [\tilde{\xi}_r(r)^2 + l(l+1)\tilde{\xi}_h(r)^2] \rho_0 r^2 dr}{M[\tilde{\xi}_r(R)^2 + l(l+1)\tilde{\xi}_h(R)^2]}, \quad (55)$$

we get E_{nl} as a normalised inertia of the mode. Define Q_{nl} as

$$Q_{nl} = \frac{E_{nl}}{\tilde{E}_0(v_{nl})}, \quad (56)$$

where $E_l(v)$ is obtained by interpolating to v in E_{nl} at fixed l , we can show that the quantity $Q_{nl} \delta v$ is independent of n and l , if the model matches the solar structure in the interior. To check if that is the case, we plot the following graph

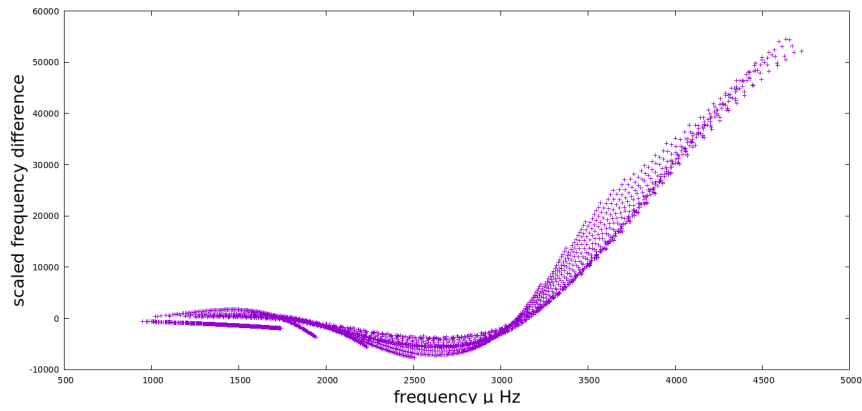


Figure 5: Variation of $Q_{nl} \delta v_{nl}$ with v_{nl}

Clearly, the variation over all the modes has collapsed into a single function, which implies that there is little difference between the model and Sun in the interior. The frequency difference arises

from the surface contribution. The surface contribution is large because firstly there are large differences between the model and the Sun near the surface as that part where convection is not efficient are difficult to model. Secondly, the sound speed is low in surface layers and hence the waves spend maximum time traversing that region. Thus the contribution gets amplified further.

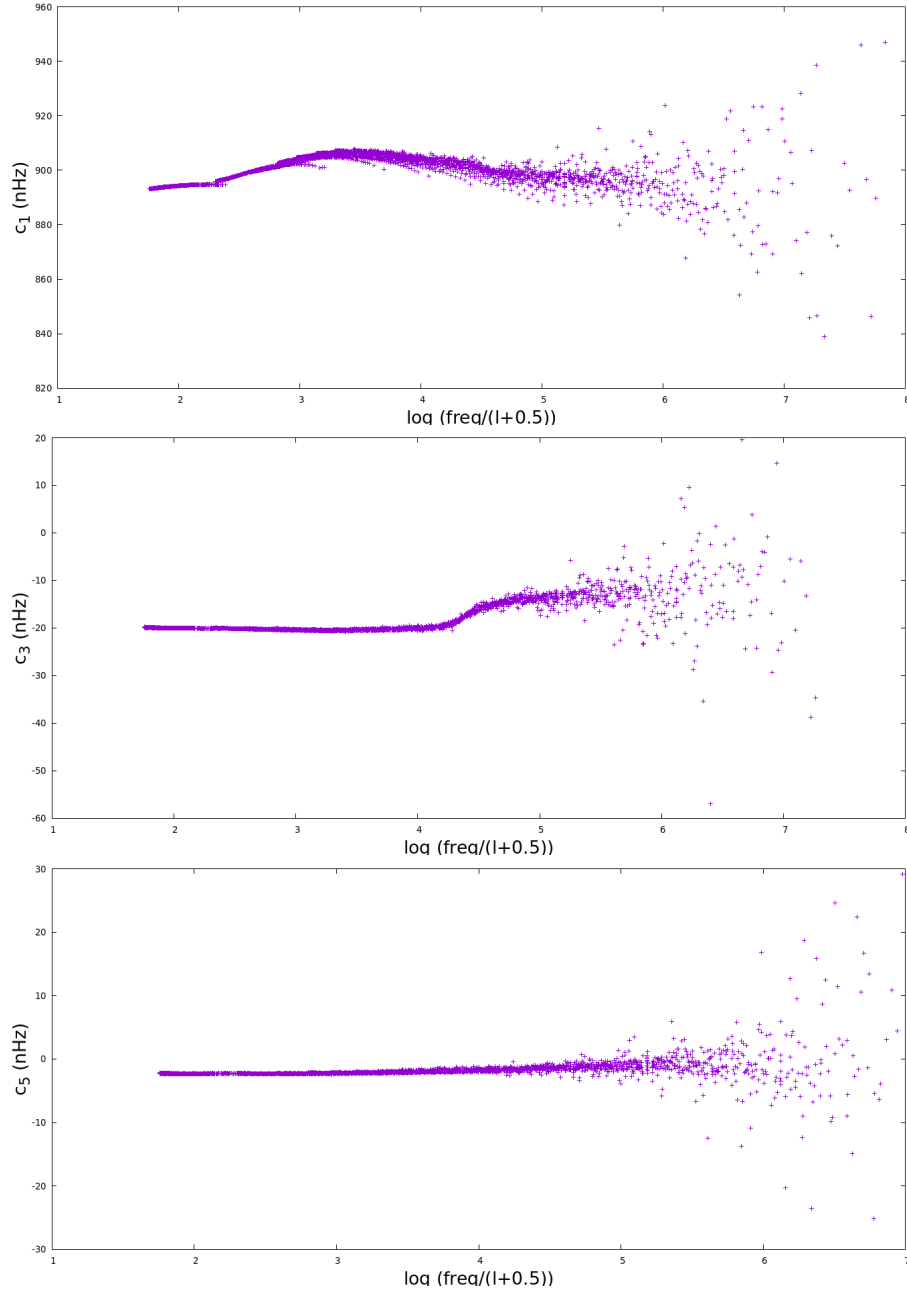


Figure 6: HMI data. Splitting coefficients c_1, c_3 and c_5 as a function of $\log(v/(l + 0.5))$, which is a measure of the penetration depth of the mode

i. Averaging kernels

Since the inversion is has been performed as a linear proces, the rotation is represented as

$$w_s(r_0) = \sum_i a_i \phi_i(r_0) \quad (57)$$

where $\phi_i(r)$ are the B-spline basis functions and a_i are the coefficients, determined from solving the inverse problem.

$$\sum_i A_{ji} a_i = \frac{c_j}{\sigma_j} \quad (58)$$

here A is the inversion matrix, c_j are the splitting coefficients and σ_j is error in splitting. The averaging kernels are defined as

$$w_s(r_0) = \int_0^{R_\odot} \mathcal{K}_{avg}(r, r_0) w_s(r) dr, \quad (59)$$

$\mathcal{K}_{avg}$ may be obtained from the above, as

$$\mathcal{K}_{avg}(r, r_0) = \sum_{j=1}^n \sum_{t=1}^N \frac{A^{-1}(t, j) \phi_t(r_0) \mathcal{K}_j}{\sigma_j} \quad (60)$$

This averaging kernel allows us to determine the correctness of the parameters used, as they are independent of the data values(c_i). The averaging kernels theoretically are expected to be peaked at r_0 , where the inversion is being calculated. The integral is also expected to be 1, since the kernels are normalised to 1

Target (r_0)	Integral $\int \mathcal{K}_{avg} dr$	Peak Value	Peak Position	Full width at half maximum	Centre of gravity $\frac{\int r \mathcal{K}_{avg} dr}{\int \mathcal{K} dr}$
0.95	1.00	38.12	0.95	3.03E-2	0.95
0.90	1.00	26.25	0.90	4.02E-2	0.90
0.85	1.00	17.45	0.86	5.94E-2	0.85
0.80	1.00	12.70	0.81	7.96E-2	0.79
0.75	1.00	9.57	0.76	0.10	0.75
0.70	1.00	7.36	0.72	0.13	0.69
0.65	1.00	6.07	0.68	0.16	0.65
0.60	1.00	5.27	0.64	0.18	0.61

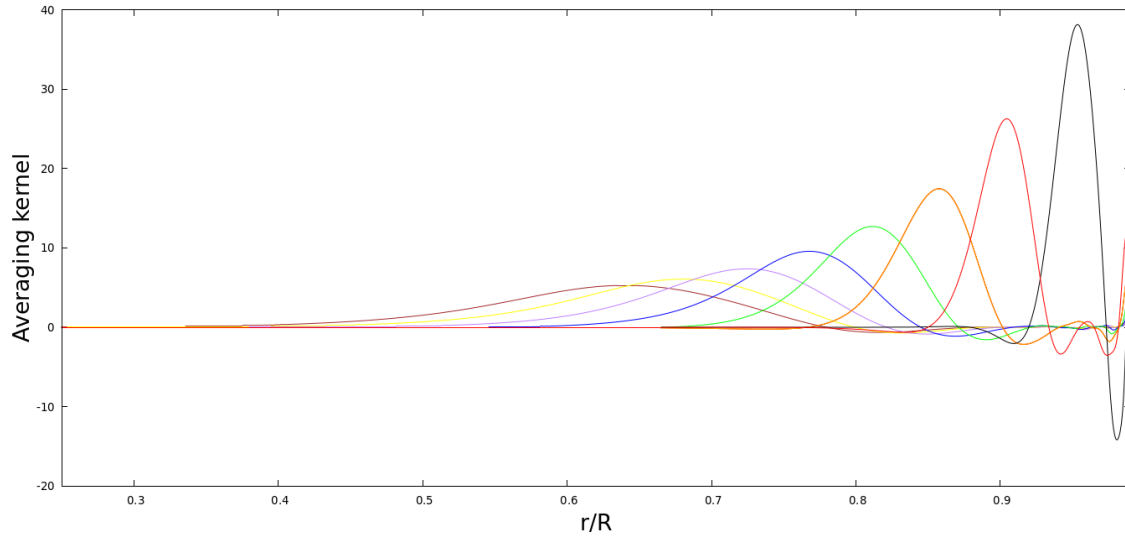


Figure 7: The averaging kernels as a function of r . The target positions are as follows: brown: 0.60, yellow: 0.65, purple: 0.70, blue: 0.75, green: 0.80, orange: 0.85, red: 0.90, black: 0.95

ii. HMI Inversion results

The splitting data from HMI were used to evaluate the velocity field of the Sun. One dimensional inversion was performed, in r , and the variation in θ was evaluated from equation 51. The smoothing parameter α were evaluated by starting with a small value and then increasing the value until the oscillations died off reasonably.

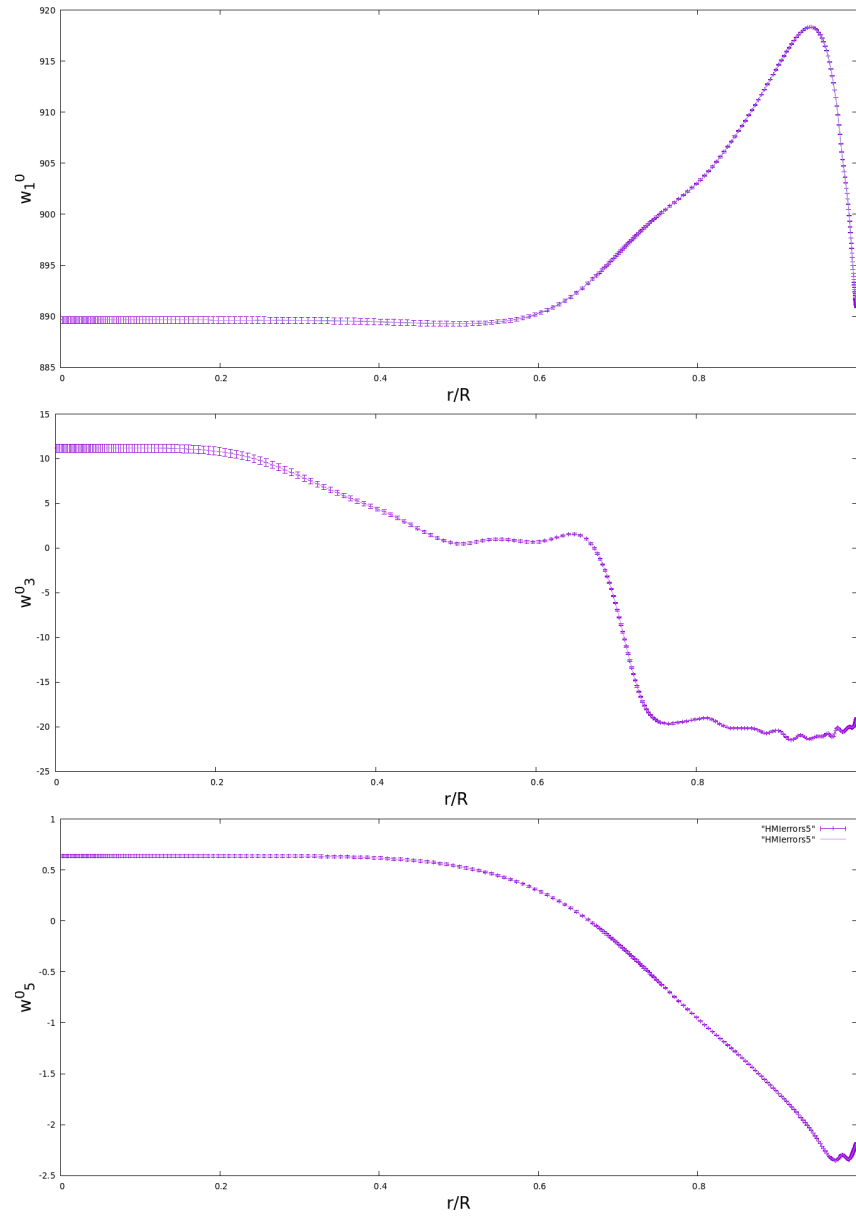


Figure 8: w_1^0, w_3^0 and w_5^0 components as a function of r . The errors in w_1^0 have been magnified 50 times

The angular velocity, thus obtained, was plotted as a polar plot, and the following image was obtained.

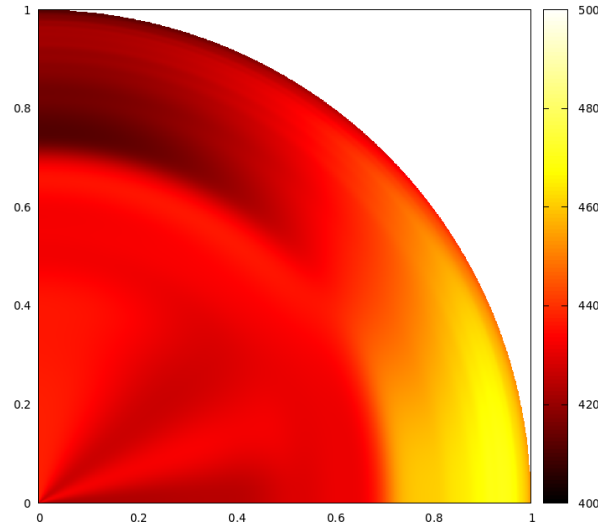


Figure 9: variation of angular velocity, in nHz

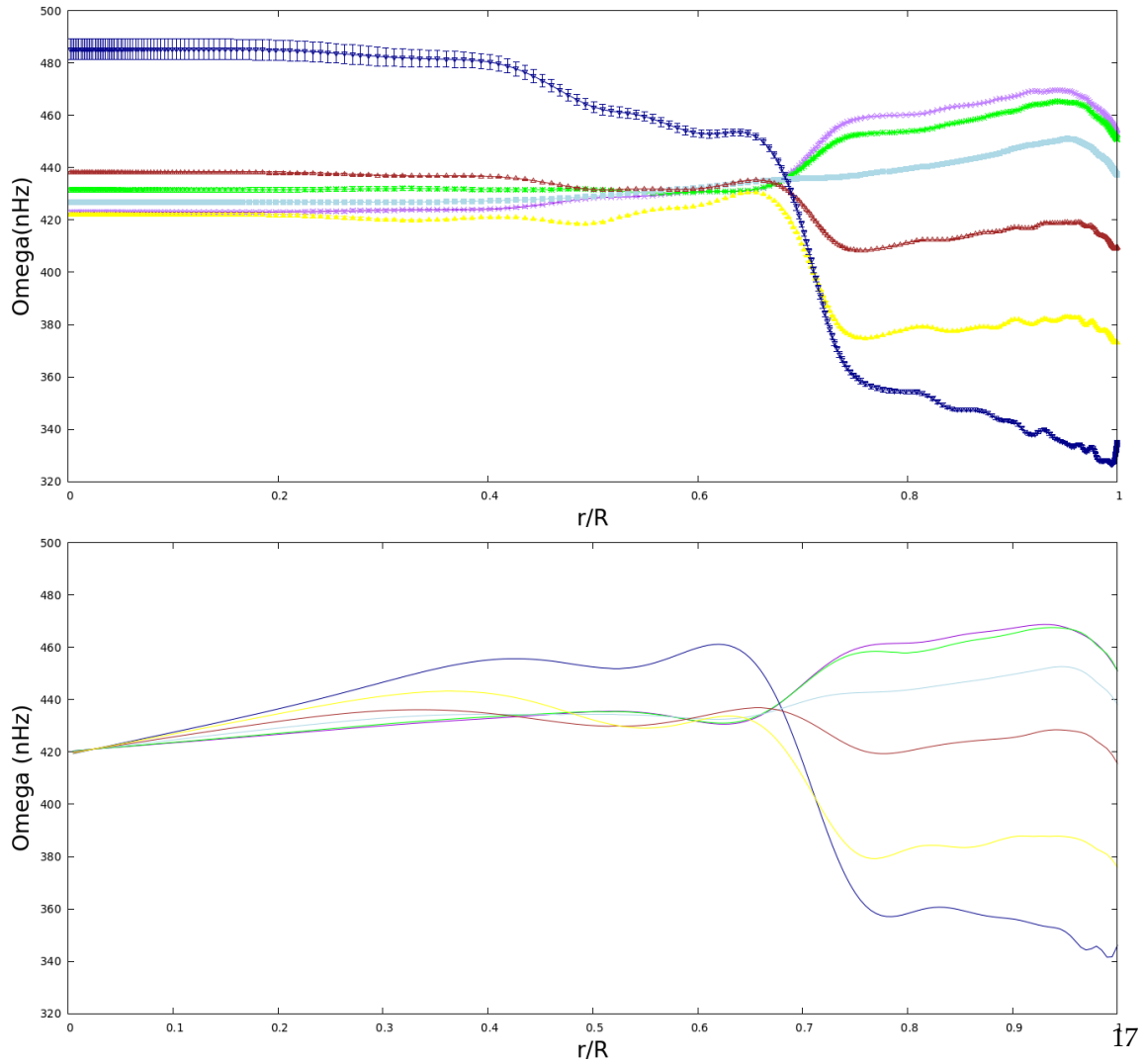


Figure 10: Variation of angular velocity, in nHz, with r at different latitudes. The top panel shows the inverted data from HMI, and the bottom panel shows the results of the 2d inversion of GONG data [3]. Purple: 0° , Green: 15° , light blue: 30° , brown: 45° , yellow: 60° , dark blue: 75° latitude.

One of the most visible, and surprising feature of the solar rotation, the tachocline, which is the sharp change in angular velocity near the base of the convection zone, is clearly visible. The shearing region near the surface, at $r > 0.95$, is also clearly visible.

iii. GONG Temporal analysis

Following this calculation, GONG data, spanning 25 years, from 1995 to 2020, were used to perform a temporal study of the variation of the velocity field of the Sun. The angular velocity for each data set, (spanning 108 days), were evaluated for the entire period. Then, the average quantities for each latitude and radius were filtered out, and the variation in the angular velocity, were plotted, as a function of time. A separate average was taken and removed for each solar cycle. Thus,

$$\delta\Omega(r, \theta, t) = \Omega(r, \theta, t) - \langle \Omega(r, \theta, t) \rangle. \quad (61)$$

The following results were obtained

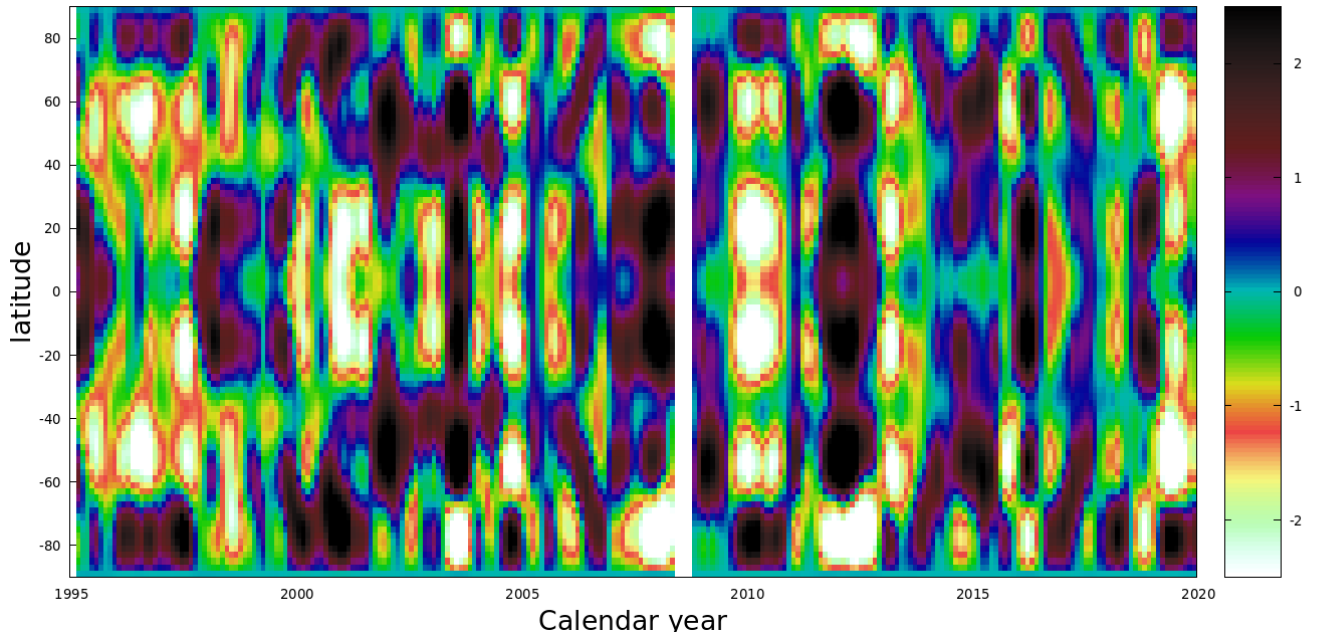


Figure 11: *variation of velocity, vs t , at $r=0.75$*

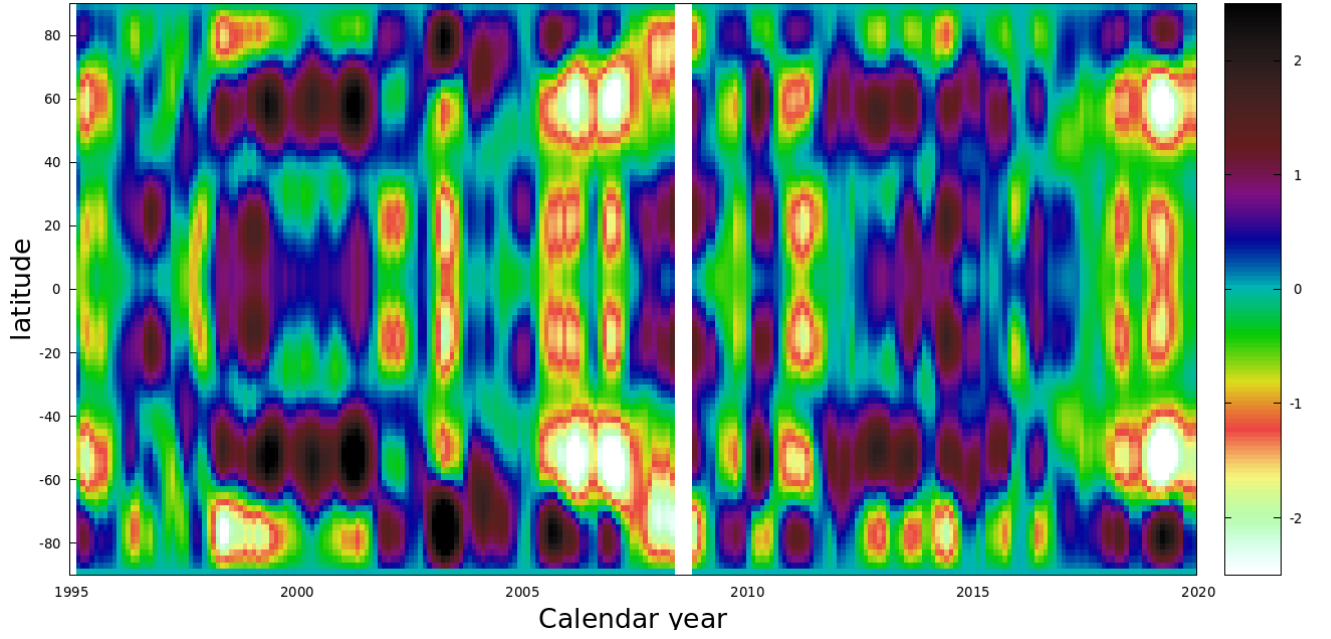


Figure 12: *variation of velocity, vs t , at $r=0.90$*

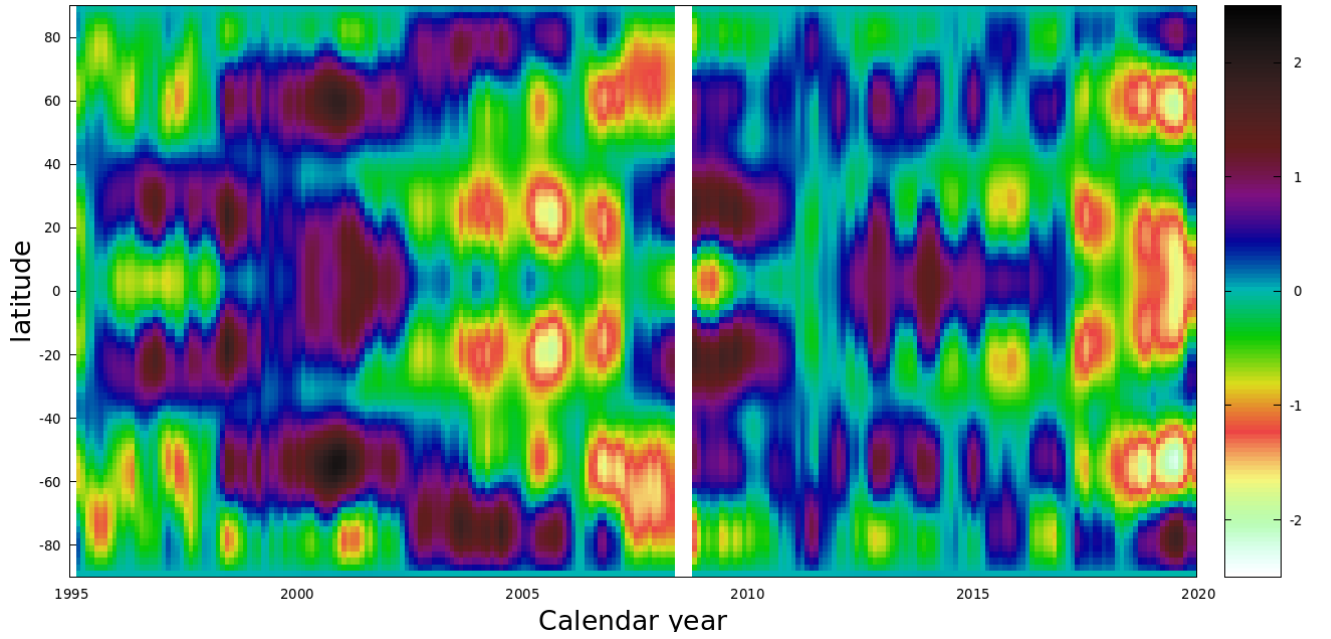


Figure 13: *variation of velocity, vs t , at $r=0.98$*

The gap in the middle is to distinguish the solar cycle 23 from the solar cycle 24. As we can see, there is a clear pattern in the variation of velocity in the outer radii.

iv. Relation to sunspots

In the figures above, the zonal flow is clearly visible, near the surface. Comparison of the pattern of temporal variations of $\delta\Omega$ at $r = 0.98R_\odot$ with the butterfly diagram shows a very strong correlation of the observed zonal flow, and the observation of sunspots. The sunspots are found to occur in regions where the magnitude of zonal flow is greater than average.

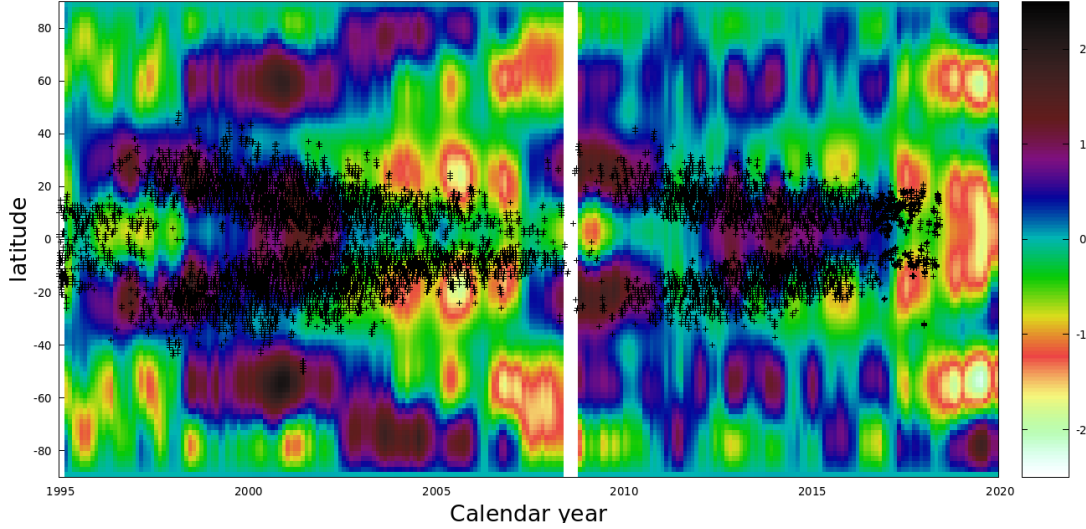


Figure 14: Variation of zonal flow with time, with the location of sunspots

VI. CONCLUSION

The study of solar oscillations presents itself as one of the very few available means of accessing the interior of the Sun. As such, its analysis presents itself with many surprising conclusions about the solar structure. The inversion of the splitting coefficients, producing the rotation velocity has revealed the unexpected tachocline, and the surface shear very prominently. The analysis of the splitting data further reveals the variation of the velocity field in the solar interior. Further study shows, that the variation can be correlated to the appearance of sunspots, and as such, present many more possibilities for revealing the details of the solar structure.

The subject of helioseismology, and the solar structure, is a very new, and very active field of research. It is mathematically, and computationally challenging, and presents many unsolved problems to uncover.

VII. ACKNOWLEDGEMENT

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